ReadSense AI — Pipeline Architecture Explained

> **What this is:** A plain-language breakdown of exactly what ReadSense does, why we use Whisper + Librosa + Gemma together, and how data flows from microphone to diagnosis. Written for the team. Last updated: 2026-07-14

🎯 What ReadSense Is Actually Trying to Do

ReadSense is a **reading diagnostics tool for children**. When a child reads a sentence out loud, a teacher cannot listen to 30 kids at once and individually diagnose *why* each one is struggling. ReadSense solves this.

It does **three distinct things**:

1. Measure *How* They Read (Acoustic Analysis)

Not just *what* they said — but *how* they said it:

- **How fast?** (Words Per Minute)
- **Where did they pause?** (and for how long?)
- **Did their voice pitch spike?** (anxiety/frustration signal)
- **Did they repeat words or stumble?** (hesitation markers)

2. Diagnose *Why* They Struggle (5 Gemma Agents)

Five specialized AI agents receive that acoustic data and reason about it:

Agent	Job
Phonetic Diagnostician	Which specific sounds/phonemes are hard? ("br" blends, "th" sounds)
Difficulty Classifier	Is the struggle phonetic, cognitive, fluency-based, or mixed?
Engagement Monitor	Is the child anxious, frustrated, or confident right now?
Practice Generator	Generate 5 personalized practice sentences targeting their weak spots
Progress Tracker	Compare today to past sessions — are they improving?

3. Report to the Teacher

A clean dashboard showing: score, struggling words, emotional state, and what to practice next. Built for teachers who have no time to manually analyze each child.

📦 What the Libraries Actually Do

These are **not** just "audio to text" tools. They extract precise diagnostic measurements.

`openai-whisper` — Word-Level Transcription with Timestamps

Whisper returns **timestamps for every single word**, not just the full transcript:

```
result = whisper.transcribe("reading.wav")

# What you actually get back:
{
  "text": "The brave dog ran quickly",
  "segments": [
    {"word": "The",     "start": 0.00, "end": 0.18, "probability": 0.99},
    {"word": "brave",   "start": 0.18, "end": 0.52, "probability": 0.97},
    {"word": "dog",     "start": 0.52, "end": 0.73, "probability": 0.99},
    {"word": "ran",     "start": 1.20, "end": 1.45, "probability": 0.95},  # ← gap before 'ran' = pause!
    {"word": "quickly","start": 1.45, "end": 1.90, "probability": 0.91},
  ]
}
```

From this alone you can calculate:

- **WPM** — `(5 words ÷ 1.90 seconds) × 60 = 158 WPM`
- **Exact pause locations** — the gap between "dog" ending at 0.73s and "ran" starting at 1.20s = **470ms pause**
- **Word confidence** — low `probability` means the child mumbled or mispronounced that specific word

`librosa` — Acoustic Signal Analysis

Whisper gives you the *linguistic* picture. Librosa gives you the *physical* sound picture:

```
What librosa measures:
├── Pitch (F0)        — Is the child's voice rising? Falling? (anxiety = rising pitch)
├── Pitch Variance    — How much does pitch fluctuate? (high variance = stress/nervousness)
├── Silence Intervals — Detect pauses whisper might miss (sub-200ms micro-hesitations)
└── Energy/Volume     — Is the child reading louder at certain words? (effort/struggle signal)
```

🔄 The Full Pipeline — Step by Step

Here is exactly what happens when a child reads a sentence:

```
Child speaks into mic
        ↓
   Browser captures audio (WebM blob via MediaRecorder API)
        ↓
   Frontend sends audio_base64 to FastAPI backend
```

```
        ↓
  ┌── WHISPER ──────────────────────────────────────────┐
  │  Input:  raw audio file                              │
  │  Output: full transcript + word-level timestamps     │
  │          + confidence scores per word                │
  └──────────────────────────────────────────────────────┘
        ↓
  ┌── LIBROSA ──────────────────────────────────────────┐
  │  Input:  raw audio file                              │
  │  Output: pitch_variance, pause_durations             │
  │          silence_intervals, energy_curve             │
  └──────────────────────────────────────────────────────┘
        ↓
  Combine into structured signal_data dict:
  {
    "transcript": "The brave dog ran quickly",
    "wpm": 95,
    "pauses": [{"at": 0.73, "duration": 0.47}],
    "pitch_variance": 340.2,
    "hesitation_count": 2,
    "word_timings": [...],
    "grade": 3,
    "target_sentence": "The brave dog ran quickly through the dark forest."
  }
        ↓
  ┌── 5 GEMMA AGENTS (run in parallel via asyncio) ─────┐
  │  All receive the structured signal_data dict         │
  │  Each reasons about it from their specialist angle   │
  │  All return clean JSON diagnosis                     │
  └──────────────────────────────────────────────────────┘
        ↓
  Dashboard renders:
  ├── Overall score (radial gauge)
  ├── Struggling phonemes ("br blend", "th")
  ├── Difficulty type (PHONETIC / FLUENCY / MIXED)
  ├── Emotional state (confident / slightly_anxious)
  ├── 5 personalized practice sentences
  └── Progress trend vs past sessions
```

✅ Why This Approach Is Better Than Feeding Raw Audio to Gemma

We tested feeding raw audio directly to `gemma-4-31b-it` and it returned:

```
400 Bad Request: Audio input modality is not enabled for this model
```

This is because the **31B Gemma model does not support audio**. Only the smaller E2B, E4B, and 12B variants do — and those are not available via the Google AI Studio API, only as local downloads.

The Doctor Analogy

Imagine you're a specialist doctor (the Gemma agent). Would you rather:

- **Option A:** Receive a chart saying: "*Patient heart rate: 142bpm, blood pressure: 150/95, temperature: 38.5°C*" → You immediately reason and diagnose.
- **Option B:** Receive a raw EKG printout and be asked to extract all the numbers yourself before diagnosing.

Whisper + Librosa are your lab instruments. They produce precise measurements. **Gemma is the specialist doctor** who receives clean, structured data and makes an expert judgment.

The 31B Gemma model is optimized for **reasoning and language understanding**, not raw audio signal processing. Giving it pre-extracted acoustic features makes its 5 diagnoses **more accurate and more reliable** than raw audio would.

🧠 Gemma Model Clarification (Important)

The blueprint has a typo (`gemma-4-9b-it` — this model does not exist). Here is the correct model lineup:

Model	Audio via API	Best For
`gemma-4-e2b-it`	❌ Local only	Mobile/Edge
`gemma-4-e4b-it`	❌ Local only	Mobile/Edge
`gemma-4-12b-it`	❌ Local only	Audio (but no API)
`gemma-4-26b-it` (MoE)	✅ AI Studio	Text reasoning
`gemma-4-31b-it` (Dense)	✅ **AI Studio**	**Text reasoning — this is our model**

Our architecture is correct: use `gemma-4-31b-it` via the API for all 5 reasoning agents, and use Whisper+Librosa locally for audio preprocessing.

🔑 Key Takeaway

> ReadSense does not ask Gemma to *listen* to a child read.
> It asks Gemma to *diagnose* a child's reading based on precise acoustic measurements.
> Whisper and Librosa are the instruments. Gemma is the expert that interprets the results.

This separation is what makes the pipeline fast, reliable, and accurate — and it's exactly what the original blueprint intended.